

SECTION A

QUESTION 1

Marks

(a) Differentiate with respect to x :

(i) e^{2x}

4

(ii) xe^{x^2}

(iii) $\sin^{-1} 3x$

(iv) $\log_e \cos^2 x$

(b) Find a primitive of

3

(i) $\frac{1}{2x+3}$

(ii) $\frac{x+2}{x^2+4}$

(c) Evaluate

3

$$\frac{\pi}{2} \int_0^{\frac{\pi}{2}} (\cos x - \sin x)^2 dx$$

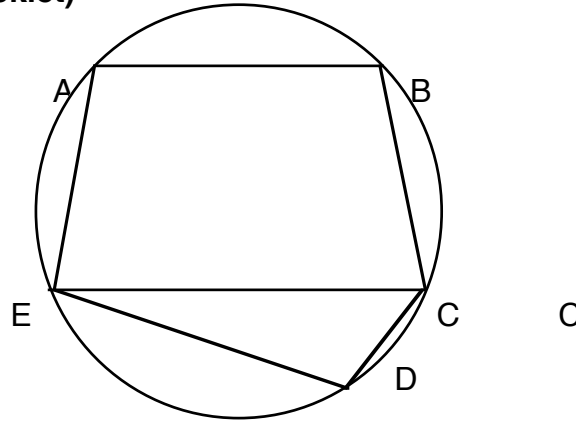
QUESTION 2**Marks**

- (a) A particle, moving in a straight line, starts from the origin with velocity 15 m/s. Its acceleration is given by $a = 7 - 4t$ where t is the time in seconds. **3**
- (i) When does the particle come to rest ?
- (ii) How far has it travelled at that time ?
- (b) A circular oil slick lies on the surface of a body of calm water. Its area is increasing at a rate of 500 m^2 per hour. At what rate is its circumference increasing when the radius of the slick is 12.5 m ? **2**
- (c) (i) Show that one root of the equation $e^x - 2\cos x - 1 = 0$ lies between 0.5 and 1 **3**
- (ii) Use Newton's Method once to find a better approximation than 0.75 for the root of the above equation.
- (d) Find the equation of the normal to the curve $y = \log_e x^2$ at the point where $x = e$ **2**

SECTION B

QUESTION 3 (Start a new booklet)

Marks



- (a) In the diagram above the chords AB and EC are parallel.

2

Copy the diagram onto your booklet.

Show that $\angle ADE = \angle BDC$

- (b) For the curve

4

$$y = (x - 2)^2 - 1$$

- (i) What is the largest positive domain for $f(x)$ to have an inverse function?
- (ii) Find this inverse function, stating its Domain and Range.
- (iii) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ on the same axes.

- (c) A particle, moving in a positive direction in a straight line, has acceleration

4

$\ddot{x} = \frac{1}{2} e^{-x} \text{ cm s}^{-2}$. Initially the velocity (v) = 1 cm s^{-1} and the displacement (x) = 0 cm .

- (i) Find the velocity of the particle when it is 2 cm from the origin.
- (ii) Find an expression for the particle's displacement with respect to time.

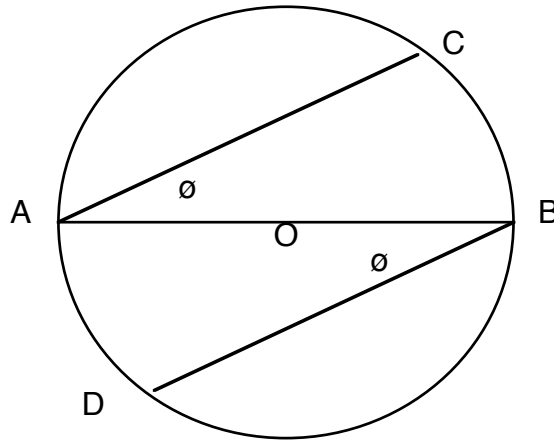
QUESTION 4

Marks

(a) (i) Show that $\tan^{-1} x + \tan^{-1} \frac{1}{x} = \frac{\pi}{2}$

1

(b)



In the diagram above AB is a diameter of a circle with centre O and radius r .

6

The chords AC and DB each make an angle θ with AB. The area between the

chords is $\frac{3}{4}$ of the area of the circle. Copy the diagram into your booklet.

(i) Show that $2\theta + \sin 2\theta = \frac{3\pi}{4}$

(ii) Join CO and OD and show that CD is a diameter of the circle.

(iii) The point C moves along the circumference at a constant rate, moving through one quadrant in 15 minutes. If A is the total area of the two triangles ACO and DBO, find, in terms of r , the rate at which A is changing when $\theta = \frac{\pi}{6}$

(iv) Find the maximum value of A and the value of θ at that time.

(c) (i) Sketch the curve $y = 3\sin 2x$ $0 \leq x \leq \frac{\pi}{2}$

3

(ii) The area between the curve $y = 3\sin 2x$, $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the x axis. Find, in terms of π , the volume of the solid generated.

SECTION C

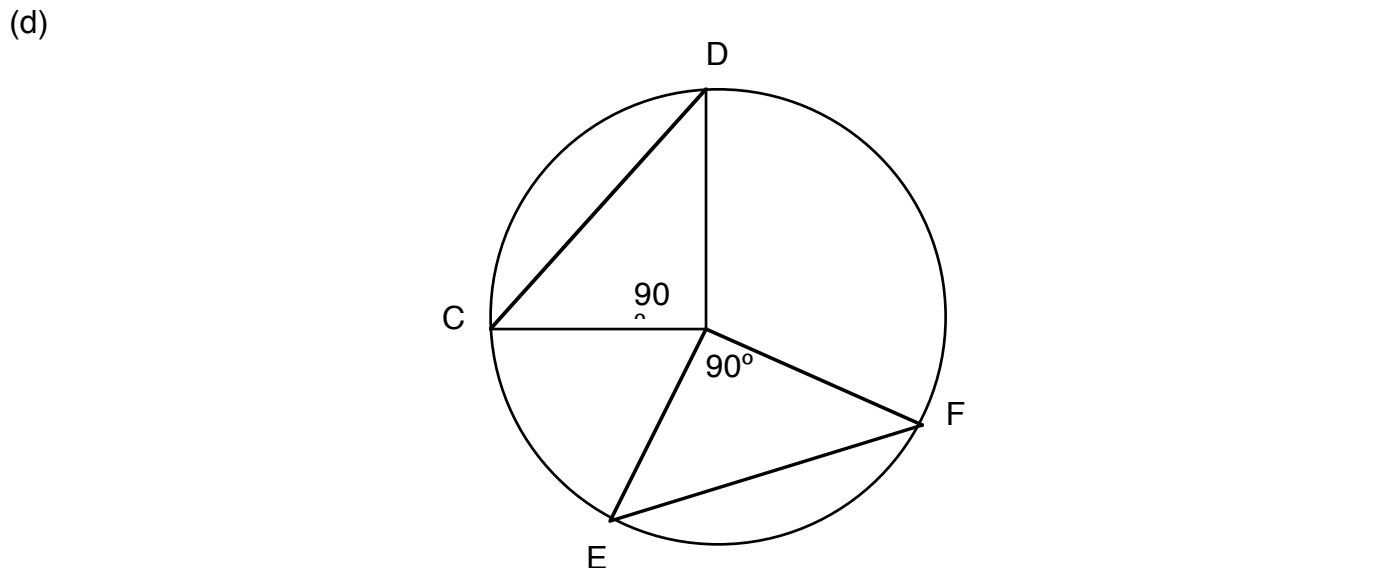
QUESTION 5 (Start a new booklet)

Marks

(a) If $\tan^{-1} x = \frac{\pi}{6}$, find the value of $\tan^{-1} \frac{2x}{1-x^2}$ **1**

(b) Find $\lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x)$ **2**

(c) Evaluate and simplify: $\frac{d}{dx} \sin^{-1} \frac{2x}{3}$ **2**



The chords CD and EF each subtend an angle of 90° at the centre of the circle. Show that the chords DE and CF meet at right angles. **2**

(e) Use the substitution $u = \tan x$ to evaluate **3**

$$\int_0^{\frac{\pi}{4}} \frac{dx}{9\cos^2 x + 25\sin^2 x}$$

QUESTION 6

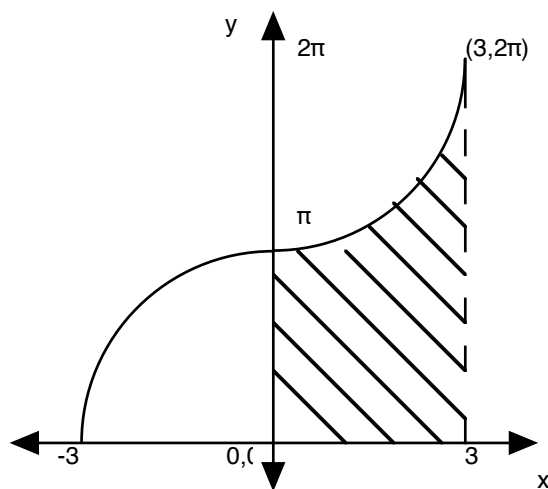
Marks

(a) Show by Mathematical Induction that

5

$$\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{4n^2 - 1} = \frac{n}{2n+1} \quad \text{for all } n \geq 1$$

(b)



The graph above is of the function $y = A \cos Bx$ where A and B are constants.

5

- Find the values of A and B.
- Find the exact area of the shaded region.



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2002

**HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 3**

Mathematics Extension 1

Sample Solutions

Question 1

$$(a) \quad (i) \quad \frac{d(e^{-2x})}{dx} = -2e^{-2x}$$

$$(ii) \quad \frac{d(xe^{x^2})}{dx} = e^{x^2} + x(2xe^{x^2}) = e^{x^2}(1+2x^2)$$

$$(iii) \quad \frac{d(\sin^{-1} 3x)}{dx} = \frac{3}{\sqrt{1-9x^2}}$$

$$(iv) \quad \ln(\cos^2 x) = 2 \ln(\cos x)$$

$$\therefore \frac{d(2 \ln(\cos x))}{dx} = 2 \cdot \frac{(-\sin x)}{\cos x} = -2 \tan x$$

$$(b) \quad (i) \quad \int \frac{1}{2x+3} dx = \frac{1}{2} \int \frac{2dx}{2x+3} = \frac{1}{2} \ln|2x+3| + C$$

$$\begin{aligned} (ii) \quad \int \frac{x+2}{x^2+4} dx &= \int \frac{x}{x^2+4} dx + \int \frac{2}{x^2+4} dx \\ &= \frac{1}{2} \int \frac{2x}{x^2+4} dx + \tan^{-1}\left(\frac{x}{2}\right) + C \\ &= \frac{1}{2} \ln(x^2+4) + \tan^{-1}\left(\frac{x}{2}\right) + C \end{aligned}$$

$$\begin{aligned} (c) \quad \int_0^{\pi/2} (\cos x - \sin x)^2 dx &= \int_0^{\pi/2} (\cos^2 x + \sin^2 x - 2 \sin x \cos x) dx \\ &= \int_0^{\pi/2} (1 - \sin 2x) dx \\ &= \left[x + \frac{1}{2} \cos 2x \right]_0^{\pi/2} \\ &= \left(\frac{\pi}{2} + \frac{1}{2} \times -1 \right) - \left(0 + \frac{1}{2} \right) \\ &= \frac{\pi}{2} - 1 \end{aligned}$$

(2) (a) $t=0, x=0, v=15$

$$a = 7 - 4t$$

(i) $v = \int a dt = 7t - 2t^2 + C$

$$\therefore v = 7t - 2t^2 + 15$$

$$v=0 \Rightarrow 2t^2 - 7t - 15 = 0$$

$$(2t+3)(t-5) = 0$$

$$\therefore \underline{t=5}$$

(ii) distance $= \int_0^5 (7t - 2t^2 + 15) dt$

$$= \left[\frac{7}{2}t^2 - \frac{2}{3}t^3 + 15t \right]_0^5$$

$$= \frac{7}{2} \times 25 - \frac{2}{3} \times 125 + 15 \times 5 - 0$$

$$=$$

(b) $A = \pi r^2$

$$\frac{dA}{dt} = 500 = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= 2\pi r \times \frac{dr}{dt}$$

$$500 = 2\pi \times 12.5 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{500}{25\pi} =$$

$$\therefore C = 2\pi r$$

$$\frac{dC}{dt} = 2\pi \frac{dr}{dt} =$$

(c) (i) $f(x) = e^x - 2\cos x - 1$, f is a continuous function

$$f(0.5) = e^{0.5} - 2\cos 0.5 - 1 \approx -1.11$$

[Remember to use RADIANS]

$$f(1) = e - 2\cos 1 - 1 \approx 0.64 > 0$$

$\therefore f(1) \times f(0.5) < 0$, so with f continuous $f(c) = 0$ with $0.5 < c < 1$

(ii) $f'(x) = e^x + 2\sin x$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.75 - \frac{e^{0.75} - 2\cos 0.75 - 1}{e^{0.75} + 2\sin 0.75}$$

$$\approx 0.85$$

(2) (d) $y = \log_e x^2 = 2 \log_e x$

$x = e \quad y = 2 \ln e = 2$

$y' = \frac{2}{x} \Rightarrow \text{at } x = e \quad y' = \frac{2}{e}$

$\therefore m_{\perp} = -\frac{e}{2}$

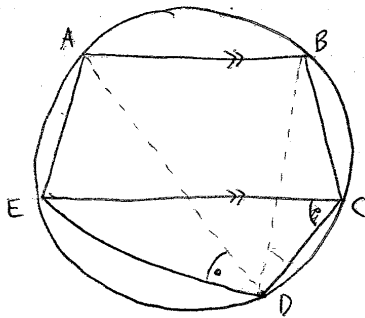
$\therefore y - 2 = -\frac{e}{2}(x - e)$

$\therefore 2y - 4 = -ex + e^2$

$\therefore ex + 2y + e^2 - 4 = 0$

(3)

(a)



ABCE is an isosceles trapezium

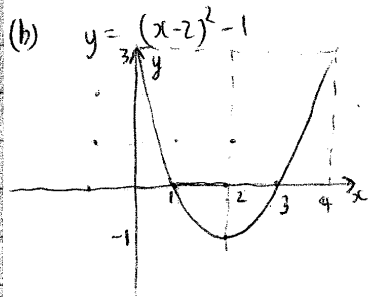
$\angle E + \angle B = 180^\circ$ (cyclic quad)

$\angle C + \angle A = 180^\circ$ (co-interior)

$\therefore \angle E = \angle C$

$\therefore AE = BC$

$\therefore \angle BDC = \angle ADE$ (equal chords subtend equal angles at the circumference)

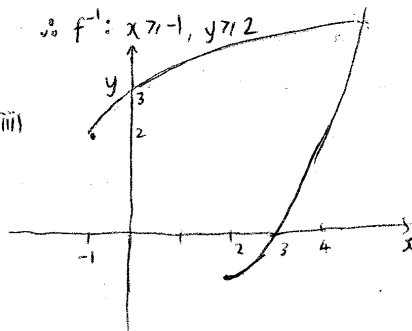


(i) $x \geq 2$

(ii) $f: x \geq 2, y \geq -1$

$\therefore f^{-1}: x \geq -1, y \geq 2$

(iii)



3 (c) $t=0, v \geq 0$

$x=0$

$a = -\frac{1}{2}e^{-x}$

$\frac{d(\frac{1}{2}v^2)}{dx} = -\frac{1}{2}e^{-x}$

$\frac{1}{2}v^2 = \frac{1}{2}e^{-x} + C$

$v^2 = e^{-x} + C_1$

$\therefore C_1 = 0 \Rightarrow v^2 = e^{-x}$

$(v \neq 0) \therefore v = e^{-x/2}$ (positive square root)

(i) $x=2 \Rightarrow v = e^{-1} = \frac{1}{e}$ cm/s

(ii) $v = e^{-x/2}$

$\therefore \frac{dx}{dt} = e^{-x/2} \Rightarrow \frac{dt}{dx} = e^{x/2}$

$t = \frac{1}{2}e^{x/2} + C_2$

$0 = \frac{1}{2} + C_2$

$\therefore t = \frac{1}{2}(e^{x/2} - 1)$

$\therefore e^{x/2} = 2t + 1$

$\therefore x/2 = \ln(2t + 1)$

$\therefore x = 2 \ln(2t + 1)$

(4) (a) For $x > 0$ $\tan^{-1}(x) + \tan^{-1}(\frac{1}{x}) = \frac{\pi}{2}$

Let $f(x) = \tan^{-1}x + \tan^{-1}\frac{1}{x}$

$\therefore f'(x) = \frac{1}{1+x^2} + \left(-\frac{1}{x^2}\right) \frac{1}{1+\frac{1}{x^2}} = \frac{1}{1+x^2} - \frac{1}{1+x^2} = 0, x > 0$

$\therefore f(x) = k$, constant

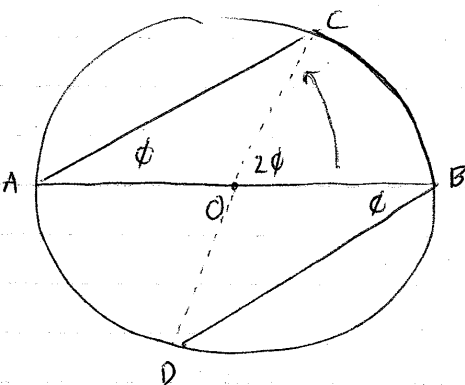
$f(1) = k$

$= \tan^{-1}(1) + \tan^{-1}(1) = 2 \times \frac{\pi}{4} = \frac{\pi}{2}$

$\therefore \tan^{-1}x + \tan^{-1}\frac{1}{x} = \frac{\pi}{2}, x > 0$

[NB if $x < 0$
 $\tan^{-1}x + \tan^{-1}\frac{1}{x} = -\frac{\pi}{2}$]

4(b)



$AC \parallel DB$

$\angle BOC = 2\phi$ (angle at centre)

$$\begin{aligned} \text{sector } BOC &= \frac{1}{2} r^2 \cdot 2\phi \\ &= r^2 \phi \end{aligned}$$

$$\begin{aligned} \Delta AOC &= \frac{1}{2} r^2 \sin(\pi - 2\phi) \\ &= \frac{1}{2} r^2 \sin 2\phi \end{aligned}$$

$$2(r^2 \phi + \frac{1}{2} r^2 \sin 2\phi) = \frac{3\pi r^2}{4}$$

$$\therefore 2\phi + \sin 2\phi = \frac{3\pi}{4} \quad \underline{Q.E.D.}$$

(ii) $\angle AOD = 2\phi$ (angle at the centre)

$\therefore \angle DOB = \pi - 2\phi$ (straight line AB)

$\therefore \angle COB + \angle DOB = \pi$

$\therefore C, O, D$ are collinear

$\therefore CD$ is a diameter.

1 quadrant in 15 min

$$\begin{aligned} \text{(iii)} \quad A &= 2 \times \frac{1}{2} r^2 \sin(\pi - 2\phi) \\ &= r^2 \sin 2\phi \end{aligned}$$

$$\begin{aligned} \frac{dA}{dt} &= r^2 \cos 2\phi \times 2 \frac{d\phi}{dt} \quad (\text{chain rule}) \\ &= 2r^2 \cos 2\phi \frac{d\phi}{dt} \end{aligned}$$

$$\angle COB = 2\phi$$

$$\therefore 2 \frac{d\phi}{dt} = \frac{\pi/2}{15} \text{ rads/min}$$

$$\therefore \frac{d\phi}{dt} = \frac{\pi}{60} \text{ rads/min}$$

$$\begin{aligned} r = \frac{\pi}{6} \Rightarrow \frac{dA}{dt} &= 2r^2 \cos \frac{\pi}{3} \cdot \frac{\pi}{60} \\ &= \frac{r^2 \pi}{60} \text{ units/min} \end{aligned}$$

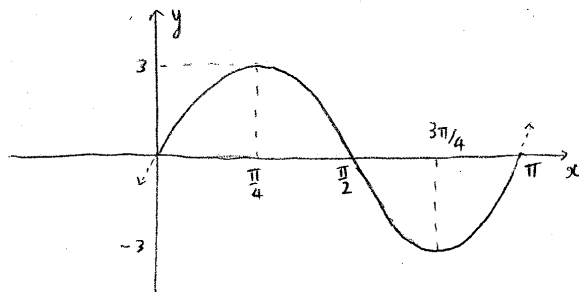
$$\begin{aligned} \text{(iv)} \quad \frac{dA}{d\phi} &= 2r^2 \cos 2\phi. \text{ So } \frac{dA}{d\phi} = 0 \Rightarrow \cos 2\phi = 0 \\ & \quad 2\phi = \frac{\pi}{2}, \frac{3\pi}{2} \\ & \quad \phi = \frac{\pi}{4}, \frac{3\pi}{4} \\ \frac{d^2A}{d\phi^2} &= -4r^2 \sin 2\phi \\ \phi = \frac{\pi}{4} \quad \frac{d^2A}{d\phi^2} &< 0 \quad \therefore \text{max} \end{aligned}$$

(4)(b) (iv) A is max at $\theta = \frac{\pi}{4}$

$$A = r^2 \sin \frac{\pi}{2} \\ = r^2 \text{ unit}^2$$

(c) (i) $y = 3 \sin 2x$

$$0 \leq x \leq \pi$$



$$(ii) V = \pi \int_0^{\pi} 9 \sin^2 2x \, dx$$

$$= \frac{9\pi}{2} \int_0^{\pi} 2 \sin^2 2x \, dx = \frac{9\pi}{2} \int_0^{\pi} (1 - \cos 4x) \, dx$$

$$= \frac{9\pi}{2} \left[x - \frac{1}{4} \sin 4x \right]_0^{\pi}$$

$$= \frac{9\pi}{2} [(\pi - 0) - (0)] = \frac{9\pi^2}{2} \text{ unit}^3$$

$$(5) (a) \tan^{-1} x = \frac{\pi}{6} \Rightarrow x = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\therefore \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \tan^{-1} \left(\frac{2/\sqrt{3}}{1 - \frac{1}{3}} \right) = \tan^{-1} \left(\frac{2/\sqrt{3}}{2/3} \right) = \tan^{-1}(\sqrt{3}) \\ = \pi/3$$

$$(b) \lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x)$$

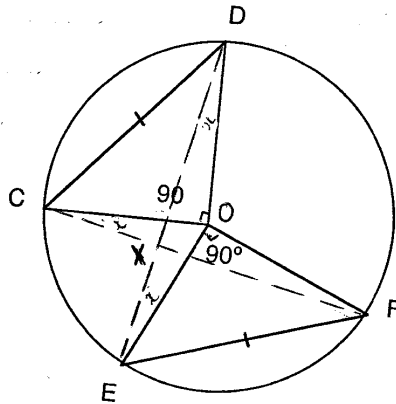
$$\begin{aligned} & \operatorname{cosec} x - \cot x \\ &= \operatorname{cosec} x - \cot x \times \frac{\operatorname{cosec} x + \cot x}{\operatorname{cosec} x + \cot x} \\ &= \frac{\operatorname{cosec}^2 x - \cot^2 x}{\operatorname{cosec} x + \cot x} = \frac{1}{\operatorname{cosec} x + \cot x} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{\operatorname{cosec} x + \frac{1}{\tan x}} \\ &= \frac{\tan x}{\tan x \operatorname{cosec} x + 1} \end{aligned}$$

$$\therefore \lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x) \\ = \lim_{x \rightarrow 0} \frac{\tan x}{\tan x \operatorname{cosec} x + 1} \\ = \frac{0}{0+1} \\ = 0$$

$$\begin{aligned}
 (5) \quad (c) \quad \frac{d}{dx} \left(\frac{\sin^{-1}(\frac{2x}{3})}{3} \right) &= \frac{2}{3} \times \frac{1}{\sqrt{1 - \frac{4x^2}{9}}} \\
 &= \frac{2}{3} \times \frac{3}{\sqrt{9 - 4x^2}} \\
 &= \frac{2}{\sqrt{9 - 4x^2}}
 \end{aligned}$$

(d)



$$\angle OCD = 45^\circ \text{ (isosceles } \Delta \text{)}$$

$$\text{Let } \angle OCF = x$$

$$\therefore \angle EDO = x \quad (\triangle ODE \cong \triangle OCF)$$

$$\therefore \angle XCD = 45 + x$$

$$\angle CDX = 45 - x$$

$$\therefore \angle DXC = 90^\circ$$

$$\therefore CF \perp DE$$

$$(e) \quad \int_0^{\pi/4} \frac{dx}{9\cos^2 x + 25\sin^2 x} \quad \div \cos^2 x$$

$$= \int_0^{\pi/4} \frac{\sec^2 x dx}{9 + 25\tan^2 x}$$

$$u = \tan x$$

$$\therefore x=0 \Rightarrow u=0$$

$$x=\pi/4 \Rightarrow u=1$$

$$= \int_0^1 \frac{du}{9 + 25u^2}$$

$$\begin{aligned}
 &= \frac{1}{25} \int_0^1 \frac{du}{\frac{9}{25} + u^2} = \frac{1}{25} \times \frac{5}{3} \tan^{-1} \left(\frac{5u}{3} \right) \Big|_0^1 \\
 &= \frac{1}{15} \tan^{-1} \left(\frac{5}{3} \right)
 \end{aligned}$$

$$\approx 0.06869$$

(6) (a)

$$\frac{1}{3} + \frac{1}{15} + \dots + \frac{1}{4n^2-1} = \frac{n}{2n+1} \quad n \geq 1$$

Test $n=1$

$$\text{LHS} = \frac{1}{3}$$

$$\text{RHS} = \frac{1}{2 \times 1 + 1} = \frac{1}{3}$$

$\therefore$ True for $n=1$

Assume true for $n=k$

$$\text{i.e. } \frac{1}{3} + \frac{1}{15} + \dots + \frac{1}{4k^2-1} = \frac{k}{2k+1} \quad \text{--- (*)}$$

NTP true for $n=k+1$

$$\text{u. } \frac{1}{3} + \frac{1}{15} + \dots + \frac{1}{4k^2-1} + \frac{1}{4(k+1)^2-1} = \frac{k+1}{2(k+1)+1} = \frac{k+1}{2k+3}$$

$$\text{LHS} = \frac{1}{3} + \frac{1}{15} + \dots + \frac{1}{4k^2-1} + \frac{1}{4(k+1)^2-1}$$

$$= \frac{k}{2k+1} + \frac{1}{4(k+1)^2-1} \quad \text{from (*)}$$

$$= \frac{k}{2k+1} + \frac{1}{(2k+3)(2k+1)}$$

$$4(k+1)^2-1 = (2(k+1)+1)(2(k+1)-1)$$

$$= \frac{k(2k+3) + 1}{(2k+1)(2k+3)}$$

$$= \frac{2k^2+3k+1}{(2k+1)(2k+3)} = \frac{(k+1)(2k+1)}{(2k+1)(2k+3)}$$

$$= \frac{k+1}{2k+3}$$

$$= \text{RHS}$$

$\therefore$ If true for $n=k \Rightarrow$ true for $n=k+1$

$\therefore$ By the principle of mathematical induction then

$$\frac{1}{3} + \frac{1}{15} + \dots + \frac{1}{4n^2-1} = \frac{n}{2n+1} \quad \text{for } n \geq 1$$

$$(6) (b) \quad y = A \cos^{-1} Bx$$

$$\cos^{-1}(0) = \frac{\pi}{2}$$

$$\therefore A \times \frac{\pi}{2} = \pi$$

$$\therefore A = 2$$

$$\Rightarrow y = 2 \cos^{-1} Bx$$

$$(-3, 0)$$

$$0 = 2 \cos^{-1}(-3B)$$

$$\therefore \cos^{-1}(-3B) = 0$$

$$-3B = \cos 0 = 1$$

$$\therefore B = -\frac{1}{3}$$

or

$$(3, 2\pi)$$

$$2\pi = 2 \cos^{-1}(3B)$$

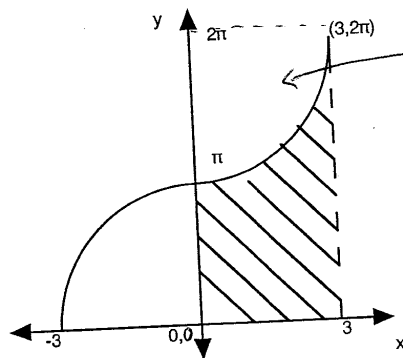
$$\pi = \cos^{-1}(3B)$$

$$3B = \cos \pi$$

$$B = -\frac{1}{3}$$

$$\boxed{\cos^{-1}(-x) = \pi - \cos^{-1}(x)}$$

$$y = 2 \cos^{-1}\left(-\frac{x}{3}\right) = 2\pi - 2 \cos^{-1}\left(\frac{x}{3}\right)$$



area

$$= \int_{\pi}^{2\pi} -3 \cos\left(\frac{x}{2}\right) dx$$

$$= \left[-6 \sin \frac{x}{2}\right]_{\pi}^{2\pi}$$

$$= -6\left(\sin \pi - \sin \frac{\pi}{2}\right)$$

$$= 6$$

$$\therefore \text{shaded area} = 6\pi - 6 \text{ sq. unit}$$